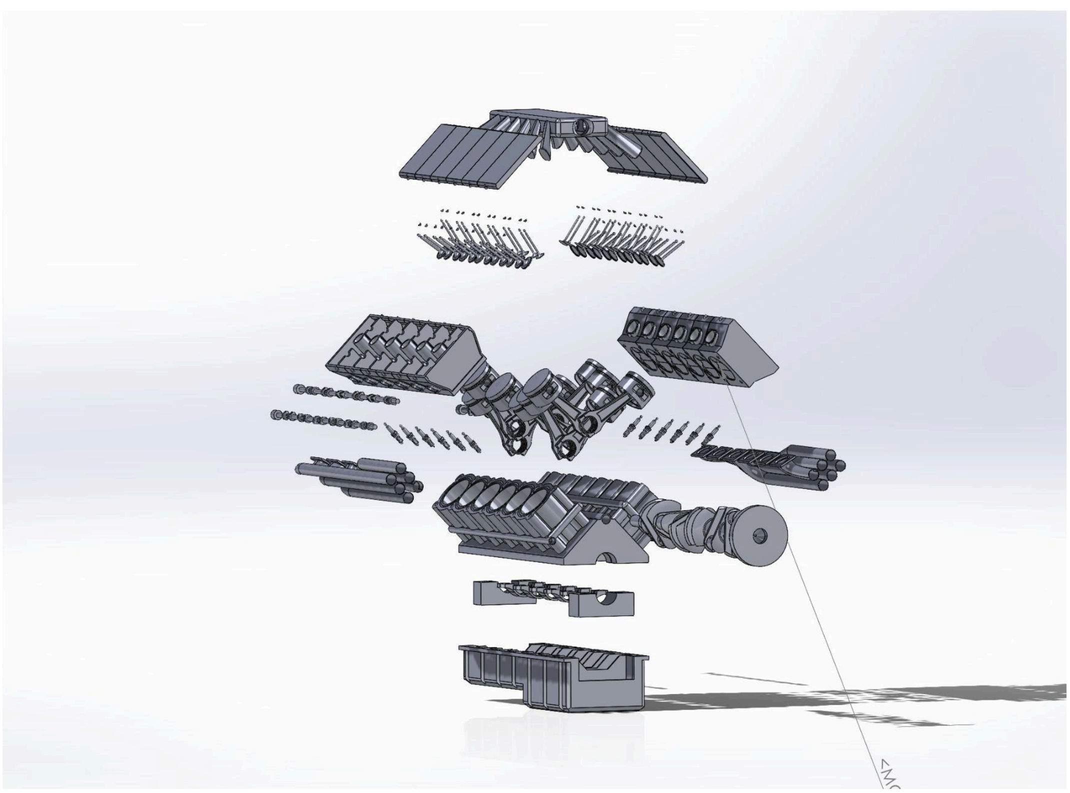
FACULTY OF ENGINEERING AND NATURAL SCIENCES

ENS 410

ADVANCED SOLID MODELING TECHNIQUES

TERM PROJECT FINAL REPORT

“V12 ENGİNE”



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Introduction:

The objective of this project was to design a detailed CAD model of a V12 internal combustion engine. The V12 engine, known for its power and smooth operation, is widely used in high-performance vehicles.This report documents the modeling process, challenges encountered, components designed, and key features of the final assembly. The project aimed to simulate the structure and function of a V12 engine through accurate modeling of essential components, such as pistons, crankshaft, connecting rods, valves, and cylinder heads. This report outlines the methodology followed for part creation and assembly, discusses design considerations, and provides an overview of the tools and features used in SolidWorks. The final model offers a clear understanding of the internal mechanics of a V12 engine.

Detailed Description of the Product:

The V12 engine that we modelled and assembled is composed of two banks of 6 cylinders in a "V" shaped arrangement and it works in 4 steps of continuous repetition: intake, compression, burst and unload.The crankshaft is driven by the pistons, which move up and down due to combustion. The camshafts control the timing of the intake and exhaust valves, allowing the air-fuel mixture in and pushing exhaust gases out. We can look at this complex structure under 7 subtitles: Engine block and core structure, crankshaft and rotating assembly, piston and connecting rod system, valve train system, cylinder head and top components, ignition and ancillary components lastly intake and exhaust system.

Engine block and core structure: The main engine block and cylinder hole block is the housing part for all the rest of the components. It has 12 cylindrical holes for pistons movement and underneath we have another crucial part "**Carter**" this component keeps oil for the pistons so that the pistons can operate as they were meant to.

Crankshaft and rotating assembly: The crankshaft is located above the pistons and connected to them. Crankshaft allows linear motion to turn into rotational motion so that pistons can move in their pattern. The pattern may change according to the structure or the type of the Engine. **The crank connection cap** secures the position of the crankshaft so that the crankshaft can work properly.

Piston and connecting rod system: There are 4 important components in this system, pistons, piston pins, connecting rods and connection caps. Pistons are modelled according to the size of the cylindrical holes of the engine. Piston pins connect pistons to connecting rods and allow pivotal motion and connecting rods transfer movement from crankshaft to pistons. Lastly connection caps secure the position of connection rods.

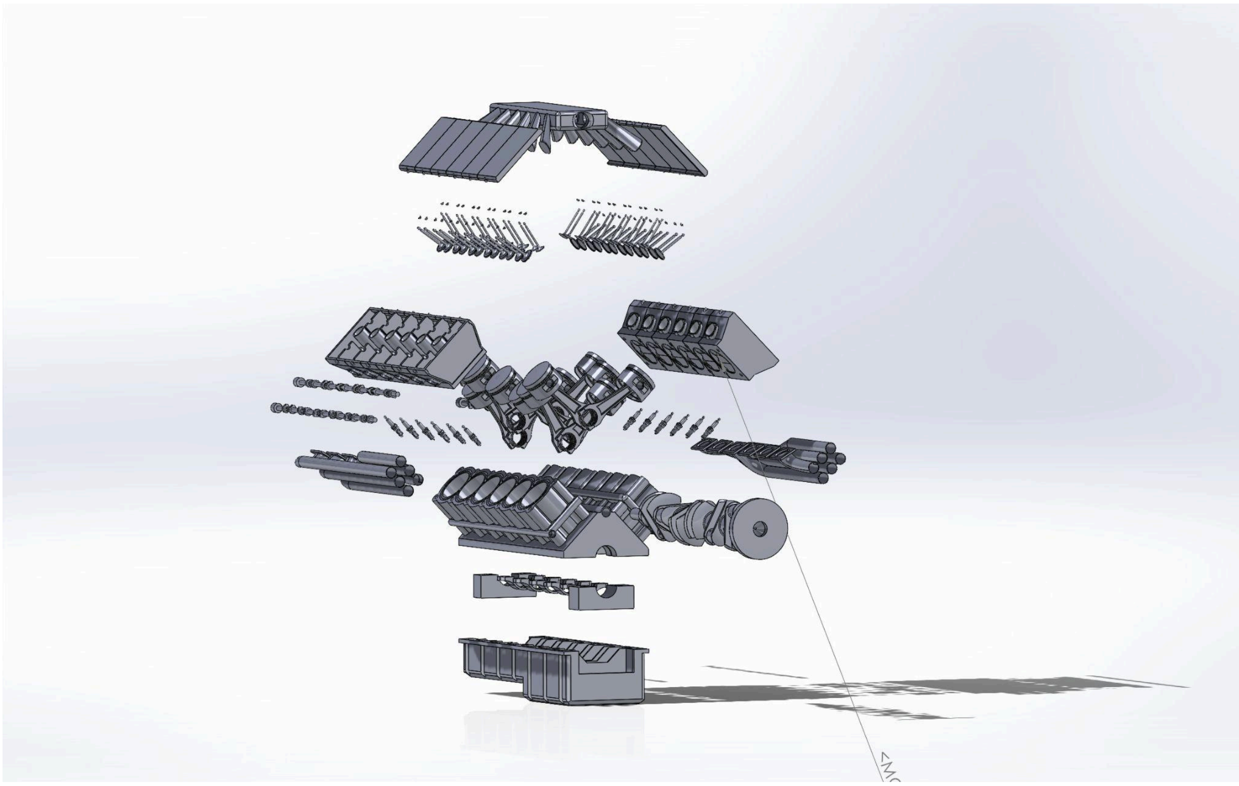
Valve train system: We have 5 subtitles for this system and these are camshaft & exhaust camshafts, Intake & exhaust valve, valve springs, valve locks (nails) and lastly Intake valve retainer & seats. Camshafts & exhaust camshafts are responsible for valve timings. Intake and exhaust valves are allowing air-gas mixture intake and exhaust explosion. Valve springs help valves to cover their position when the force is being applied around the valve stems. Valve nails prevent valves from dropping. Lastly Intake valve retainer & seats holds the springs under tension and ensures proper sealing during combustion.

Cylinder head and top components: Cylinder heads house upper components of the combustion chamber, mirrored version allows “V” shape and top cover protects camshaft and also keeps oil in control.

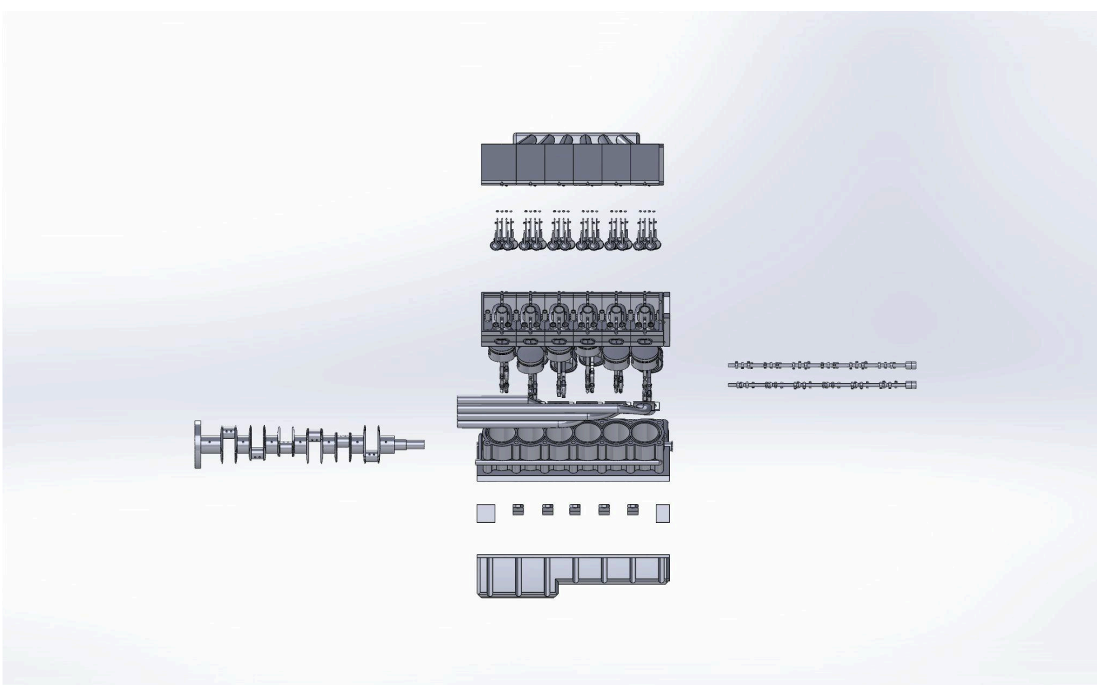
Ignition and ancillary components: It is composed of spark plugs and nails. Spark Plugs are positioned above the combustion chamber and allow ignition by giving a spark to the air fuel mixture and nails are used for fastening the other components with a tight fit.

Intake and exhaust system: This system is made up of exhaust manifolds, pipes and retainers.The manifold and pipes allow burnt gas to get out of the system so that the combustion cycle can continue and retainers are securing various valve and spring components.

Dissection Plan:



Assembly Edge view



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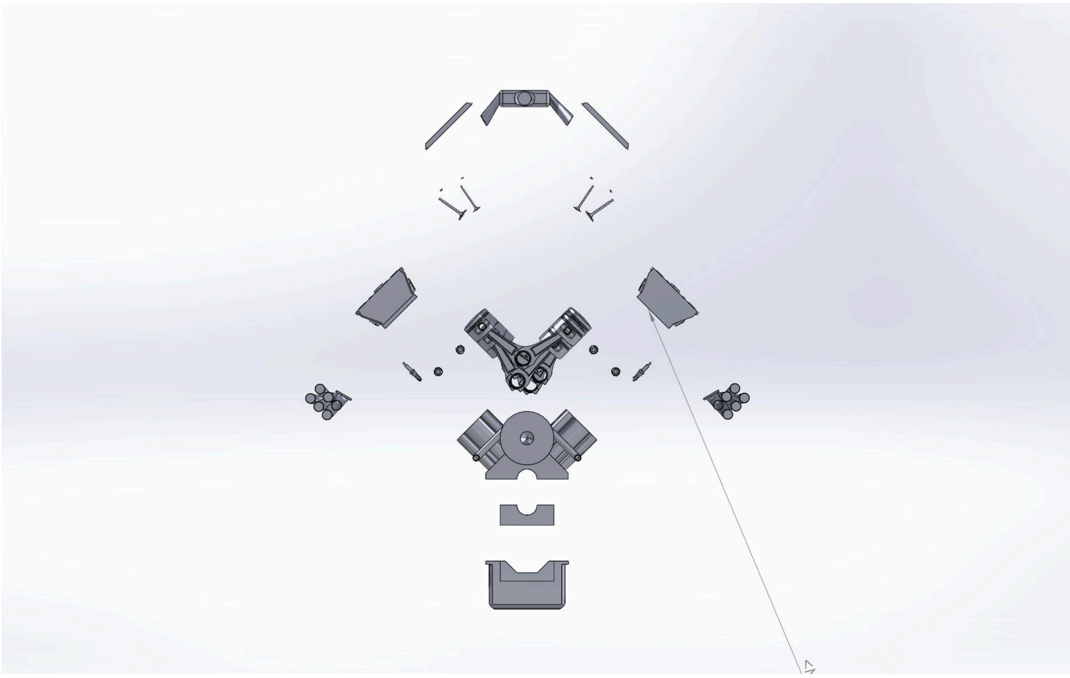
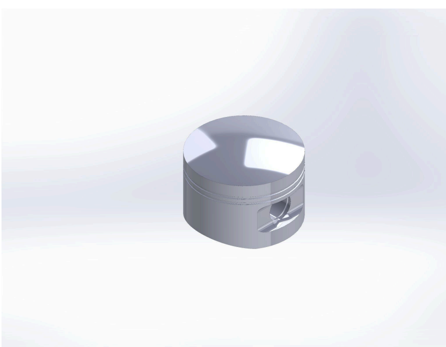
Assembly Right View

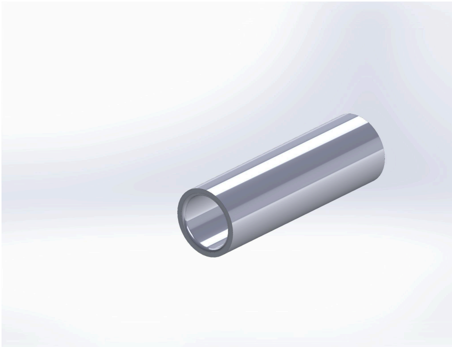


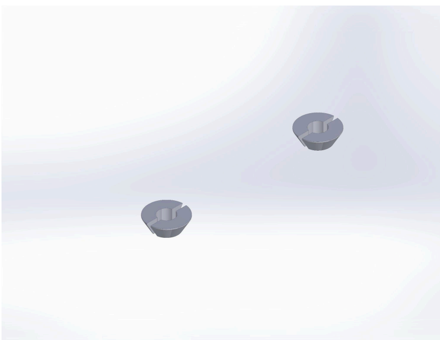
Assembly Front View

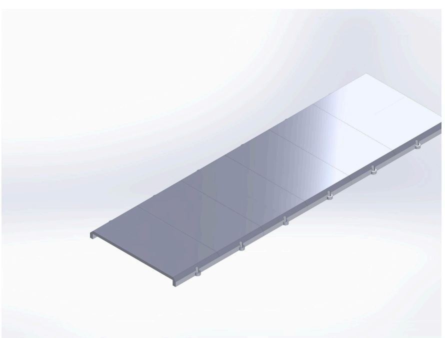


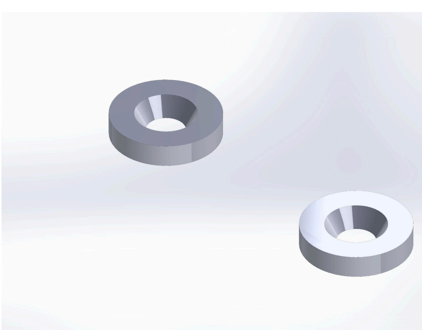
Piston / Material:AISI 4130 Steel, normalized

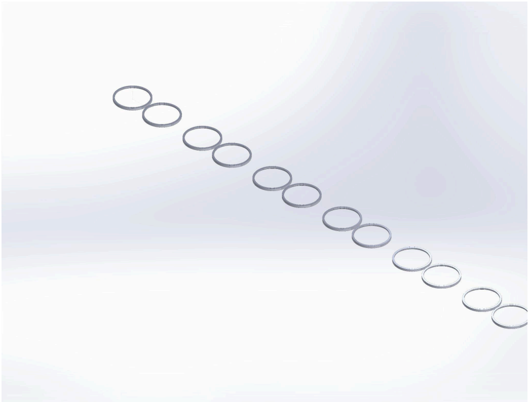
Connecting Rod P1 / Material: AISI 4340 Steel, normalized

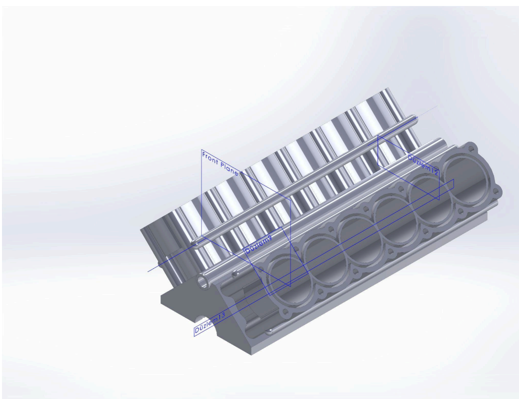

Piston Pin / Material: AISI 4130 Steel, normalized

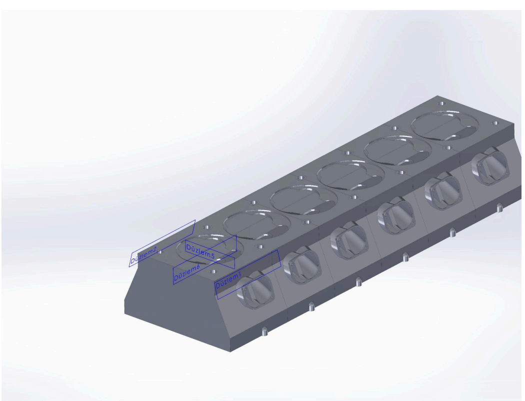

Intake and Exhaust Valve Nails / Material: AISI 4130 Steel, normalized

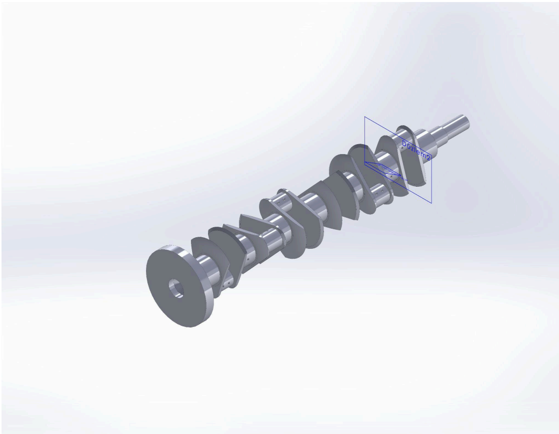

Cylinder Head Cover / Material: (Aluminum recommended)


Intake Valve Retainer / Material: AISI Type A2 Tool Steel

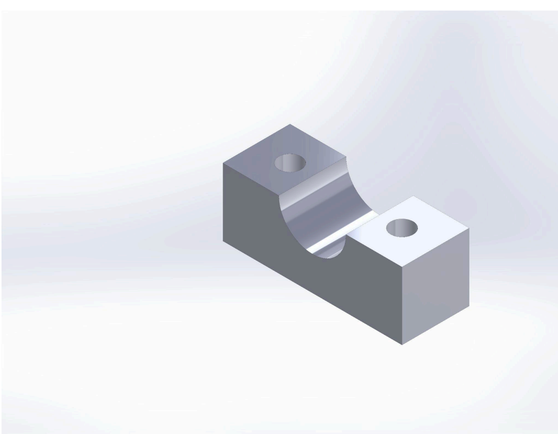

Intake Valve Seat / Material: AISI Type A2 Tool Steel

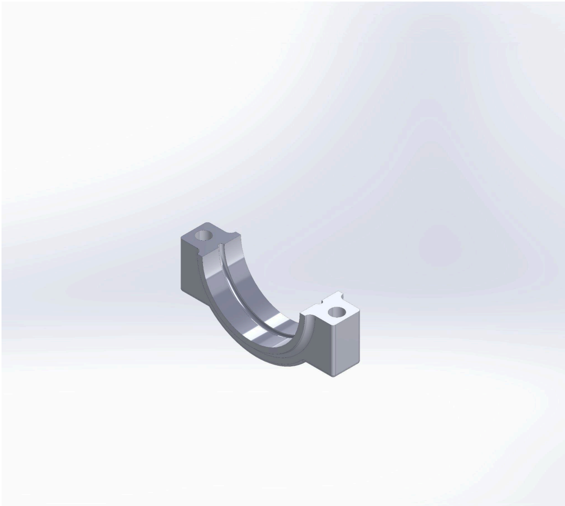

Engine Block/Cylinder Crankcase / Material: Cast Carbon Steel

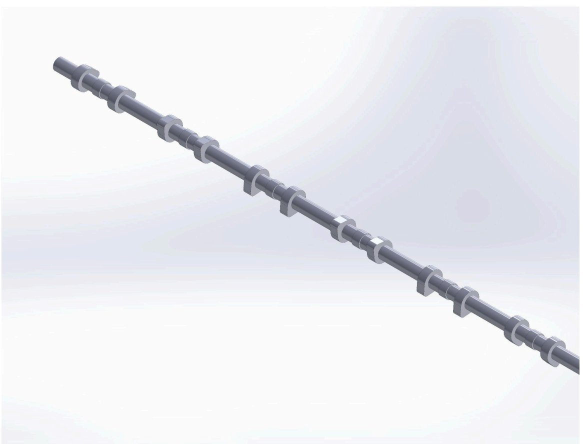

Cylinder Head / Material:(Aluminum recommended)

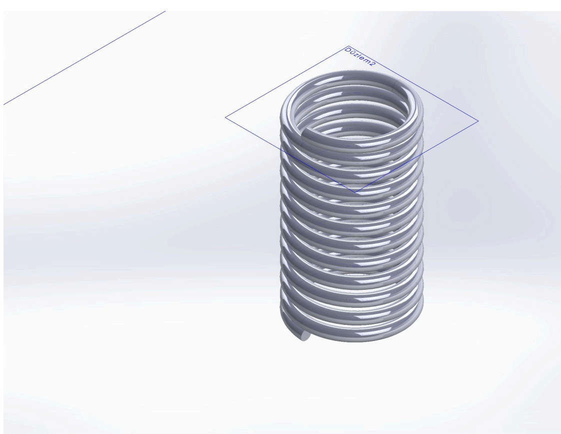

Crankshaft / Material: AISI 4340 Steel, normalized


Connecting Rod P2 / Material / AISI 4340 Steel, normalized


Connecting P3/ Material: AISI 4340 Steel, normalized

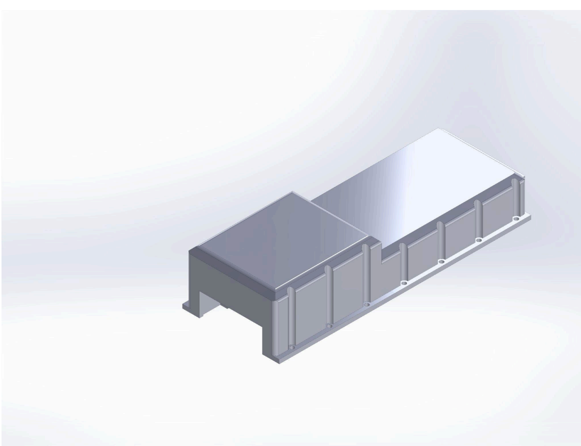

Crank Connection Part 2 / Material: AISI 4340 Steel

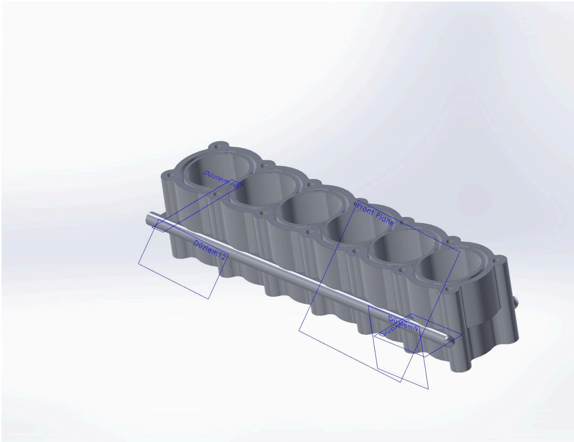

Camshaft / Material: AISI 4340 Steel, normalized

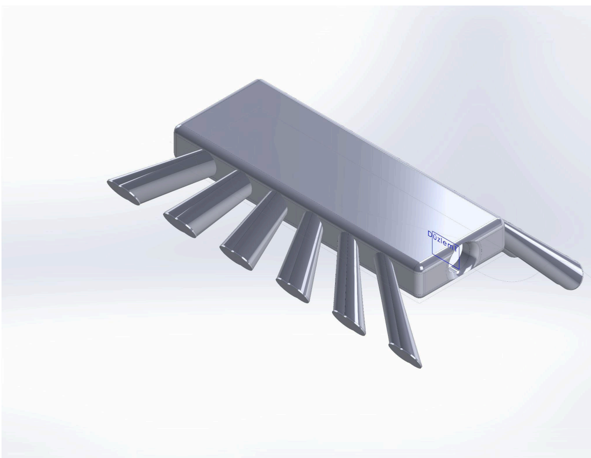

Valve springs / Material: AISI Type A2 Tool Steel

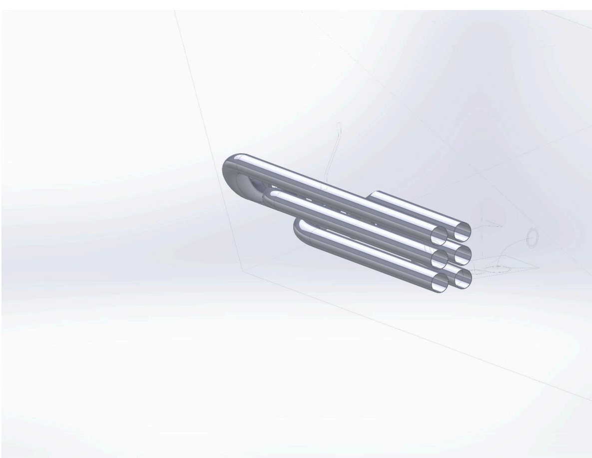

Spark Plug


Intake Valve / Material: Chrome Stainless Steel


Carter / Material: AISI 4340 Steel, normalized


Cylinder Holes / Material: Cast Carbon Steel


Exhaust Manifold /Material: Stainless Steel


Exhaust / Material: Stainless Steel

Design Discussion:

Preparing a CAD model of a V12 engine gave us a better understanding of how an engine works. During drawing parts of the engine bose and cut extrusion, swept surface, inserting helix, loft and lastly mirroring commands took most of the process. We were very careful about the dimensions of the parts that we were drawing to not have any fitment problems during assembly. Mates and constraints were added in assembly so that everything can be aligned and functional.

Overall the process required planning, precision and most importantly teamwork to finish the V12 engine.

Planned Work Distribution/ Time Table:

	SUBJECT	Time (week)	Start date	Finish Date	Person in Charge	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Choosing the design	5	1	5	ALL	█	█	█	█	█									
2	CAD model draft designing process	5	8	12	ALL								█	█	█	█	█		
3	Designing parts on solidworks	3	12	14	ALL												█	█	█
4	Making Assembly on solidworks	2	13	14	BURAK/ EGE													█	█
5	Preparing final report	1	14	14	ALP														█